Co-Evolutionary Feature Compression for Early-Stage Chronic Kidney Disease Detection on MIMIC-IV

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***Abstract*—chronic kidney disease (CKD) is a serious health problem that affects millions of people. Finding it early is hard because most hospitals need many lab tests, which are expensive. In this paper, we present SGPO v2, a method that selects the most important features and tunes the classifier at the same time. It uses three algorithms together: SFOA for feature selection, DOA for tuning, and FGO to avoid getting stuck. We tested it on the MIMIC-IV dataset with 57,875 patients. Our method reduced the features from 42 to only 8 (an 81% drop) and still got an AUC of 0.9537 and sensitivity of 0.8902. The interesting part is that it picked administrative features like insurance and marital status instead of extra lab tests. This makes CKD screening cheaper and good for hospitals with limited resources.**

***Keywords*—Chronic Kidney Disease, Feature Selection, Co-Evolutionary Optimization, MIMIC-IV, Machine Learning.**

I. INTRODUCTION

Chronic Kidney Disease (CKD) affects around 850 million people in the world [1]. It is dangerous because it grows slowly without clear symptoms until it is too late. To check for CKD, doctors usually order a full lab panel with more than 20 tests, which is expensive and not always possible in small clinics or poor areas [2].

Machine learning (ML) can help detect CKD earlier [3]. But many of the existing studies have three main problems:

1) They use small datasets like UCI CKD (only 400 patients) [4], so the results do not work well on real hospital data.

2) They apply SMOTE before splitting the data, which causes data leakage and gives unfair results [5].

3) They treat feature selection and hyperparameter tuning as two separate steps. But these two things depend on each other, so doing them separately is not optimal.

To solve these problems, we propose SGPO v2. It combines three nature-inspired algorithms in one loop. SFOA handles feature selection, DOA tunes the classifier, and FGO adds random moves when the search gets stuck. All three share one fitness matrix so they evolve together. Fig. 1 shows the overall structure.

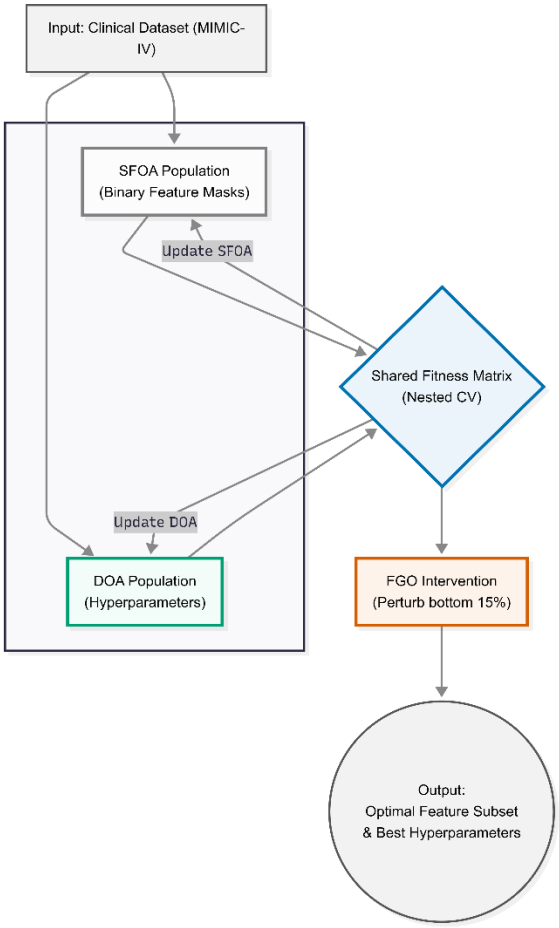

FIGURE 1: *A simple block diagram showing how SFOA, DOA, and FGO are connected through a shared fitness matrix in the SGPO v2 framework.*

Our main contributions are:

• We build a joint optimization method that does feature selection and tuning together, not in sequence.

• We reduce the number of features by 81% (from 42 to 8) while keeping the AUC at 0.9537.

• We use a strict nested cross-validation with SMOTE only in training folds, which prevents data leakage.

II. RELATED WORK

A. ML for CKD Prediction

Early work used simple models like Logistic Regression and SVM on the UCI dataset [3]. They got high accuracy (above 95%), but this was on only 400 samples so it cannot be trusted for real hospitals. Newer work uses MIMIC-IV [4] and deep learning. For example, Leung et al. [13] got an AUC of 0.91 using CNN-LSTM. The main issue with deep learning is that it is a black box. Doctors cannot see why the model made its decision, which makes them not trust it.

B. Feature Selection Methods

Feature selection methods are usually grouped into three types. Filter methods like Mutual Information are fast but ignore how features interact. Wrapper methods like RFE are better but slow. Embedded methods like Lasso do selection while training [6]. The problem with all of them is that they select features without thinking about the classifier settings.

C. Metaheuristic Algorithms

Algorithms inspired by nature are popular for feature selection. GWO [10] and PSO [11] are well-known. Newer ones like SFOA [7], DOA [8], and FGO [9] are promising. But most studies only use them for feature selection, not for tuning at the same time. Our work is different because we use all three in one co-evolutionary loop.

D. Research Gap

From the literature, we see three gaps: most work uses small datasets, does not combine feature selection with tuning, and has weak validation protocols. SGPO v2 fills these gaps. Table I compares our work with recent studies.

TABLE I. COMPARISON OF RECENT CKD STUDIES

Study	Dataset	Method	Joint Opt?	CV Protocol
Irianto [1]	UCI (400)	RF + PSO	No	K-Fold
Aswathy [9]	UCI (400)	FPA-DNN	No	N/A
Leung [13]	Hospital EHR	CNN-LSTM	No	Temporal
Chen [19]	Asian Cohort	XGBoost+SHAP	No	Cross-Val
SGPO v2	**MIMIC-IV (57,875)**	**SFOA+DOA+FGO**	**Yes**	**Nested 10×3**

III. PROBLEM FORMULATION

Given a dataset X with N patients and D features, and labels y in {0, 1}, we want to find the best feature mask s and the best hyperparameter vector θ that maximize this fitness function:

$$F(s, \theta) = w1 * AUC + w2 * Sens - w3 * (|s| / D) \quad (1)$$

Here w1, w2, w3 are weights. AUC measures how well the model separates classes. Sens is the sensitivity (true positive rate). The last term |s|/D punishes models that use too many features. We set w = 0.60, w2 = 0.30, and w3 = 0.10 because AUC is most important, then sensitivity (we do not want to miss sick patients), and finally feature count.

The constraints are:

- s_d is 0 or 1 for each feature d.
- |s| ≤ K, where K is the maximum number of features we want.
- θ must stay in the valid range [θ_min, θ_max].

IV. DATASET AND METHODOLOGY

A. MIMIC-IV Dataset

MIMIC-IV is a public EHR dataset from Beth Israel Deaconess Medical Center in Boston [4]. It has data on more than 180,000 hospital stays between 2008 and 2019. We used 57,875 adult patients (age ≥ 18) with complete renal lab data. CKD labels came from ICD-10 codes N18.x. The dataset was almost balanced (50.5% positive cases), so we did not need heavy oversampling.

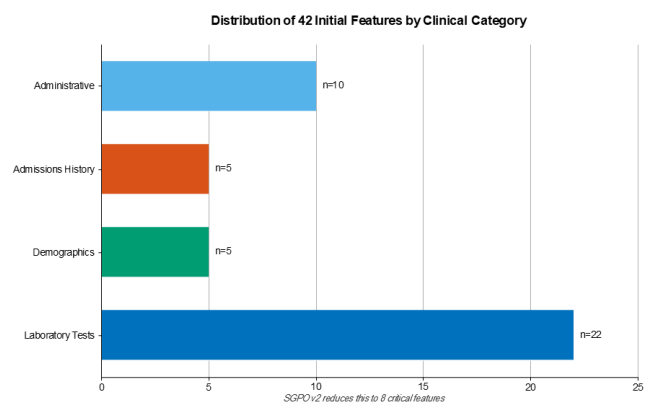

FIGURE 2: chart showing the 42 initial features grouped by category: lab tests, demographics, admissions, and administrative features.]

B. Preprocessing

We started with 42 features in six groups: renal labs, blood counts, vital signs, demographics, admin data, and admission history. We filled missing values using the median from the training set only. Categorical features were one-hot encoded. All numeric features were scaled to zero mean and unit variance using training statistics.

C. Nested Cross-Validation with Safe SMOTE

We used nested 10-fold CV. The outer loop splits the data into 10 folds and evaluates the model. Each outer fold uses an inner 3-fold loop for model selection. SMOTE was applied only inside the inner training folds, never on the test or validation data. This is important to avoid data leakage. A

common mistake in other papers is to apply SMOTE to the full dataset, which gives better-looking results but is not real. Fig. 3 shows our protocol.

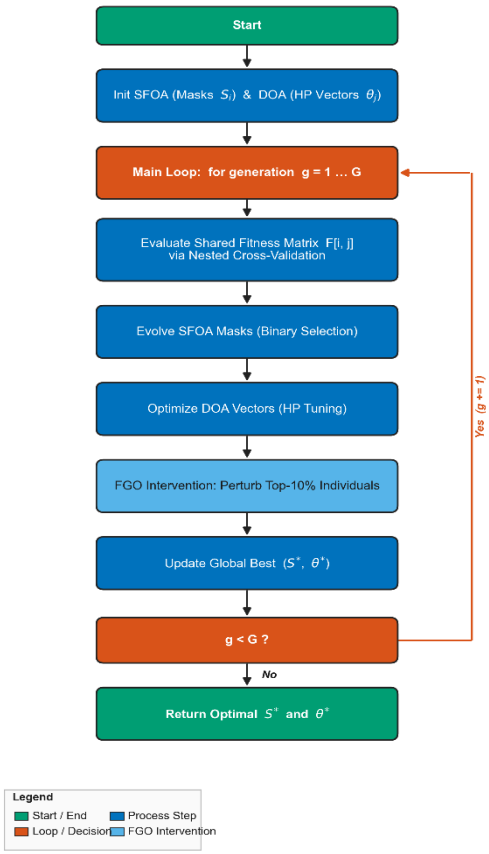

Fig. 3. flowchart showing the nested cross-validation: outer 10-fold loop, inner 3-fold loop, and where SMOTE is applied (only inside inner training folds).

V. PROPOSED FRAMEWORK: SGPO V2

A. The Three Algorithms

SGPO v2 combines three metaheuristics. SFOA [7] is inspired by starfish that can regenerate lost arms. It is good for binary search, so we use it to pick features (feature = 1 means selected). DOA [8] is inspired by the sleep cycles. It searches the continuous space of hyperparameters. FGO [9] is inspired by how fungi spread through soil. It adds random moves when the population is stuck in one place.

B. Shared Fitness Matrix

The key idea of SGPO v2 is the shared fitness matrix F of size N × M, where N is the SFOA population size and M is the DOA population size. Every feature mask is tested with every hyperparameter vector. Each cell F[i, j] stores the fitness of that pair. This way, feature selection and tuning happen together, not one after the other.

The best SFOA agent is the one with the highest score across the row:

$$F_S(s_i) = \max_j F[i, j] \quad (2)$$

Similarly, the best DOA agent is found across the column:

$$F_D(\theta_j) = \max_i F[i, j] \quad (3)$$

C. Pseudocode

Algorithm 1 shows the full SGPO v2 procedure.

Algorithm 1: SGPO v2 Co-Evolutionary Optimization

Input: Dataset X, labels y, generations G, sizes N, M, weights w.
Output: Best mask S*, best hyperparameters θ*.

```
Initialize SFOA population (feature masks)
Initialize DOA population (hyperparameters)
for g = 1 to G do
   for i = 1 to N do
      for j = 1 to M do
         Evaluate pair (s_i, θ_j) with nested CV
         F[i,j] = w1*AUC + w2*Sens - w3*|s_i|/D
      end for
   end for
   Update SFOA using best column match
   Update DOA using best row match
   Apply FGO on bottom 15% of population
end for
return best (S*, θ*) from F
```

Each generation has three phases. First, we evaluate all N × M pairs. Second, we update each population using the best match from the other. Third, FGO perturbs the weakest 15% to avoid local optima. Fig. 4 shows the full flowchart.

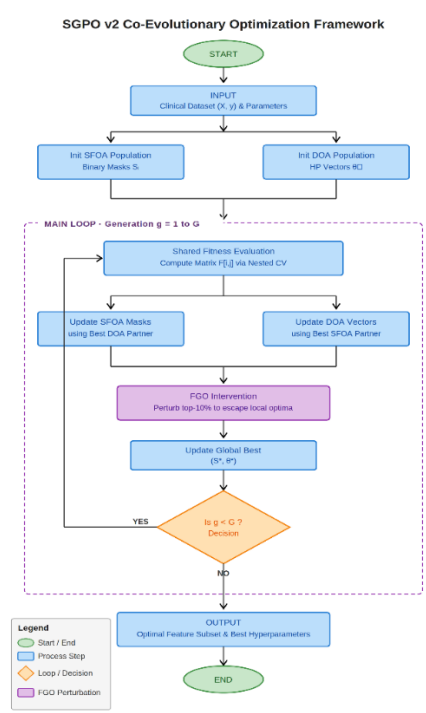

FIGURE 4: *A flowchart of the SGPO v2 co-evolutionary loop showing the three phases: fitness evaluation, co-adaptive updates, and FGO perturbation.*

VI. EXPERIMENTS AND RESULTS

A. Setup

We used SFOA population N = 10, DOA population M = 10, and G = 30 generations. The base classifier is XGBoost [13]. The hyperparameter ranges were: learning rate [0.01, 0.3], max depth [3, 10], estimators [50, 500], and L2 regularization [0, 5]. All experiments ran on a 16-core CPU with 64 GB RAM.

B. Comparison with Baselines

We compared SGPO v2 with four classical feature selection methods, all selecting exactly 8 features. Table II shows the results.

TABLE II. BASELINE COMPARISON (TOP-8 FEATURES)

Method	AUC-ROC	Sensitivity	Lab Tests
SGPO v2 (Ours)	**0.9537 ± 0.0020**	**0.8902 ± 0.0063**	**1**
RF-Importance	0.9529 ± 0.0018	0.8899 ± 0.0051	4
RFE-RF	0.9489 ± 0.0023	0.8862 ± 0.0056	3
Mutual-Info	0.9457 ± 0.0023	0.8848 ± 0.0062	3
L1-LR	0.9441 ± 0.0019	0.8770 ± 0.0046	3

SGPO v2 got the highest AUC and sensitivity. The important thing is not the small AUC difference, but the feature set. Classical methods all picked extra lab tests (Urea, Hemoglobin, RBC). Our method picked only one lab test (creatinine) plus cheap administrative features (insurance, marital status, admissions).

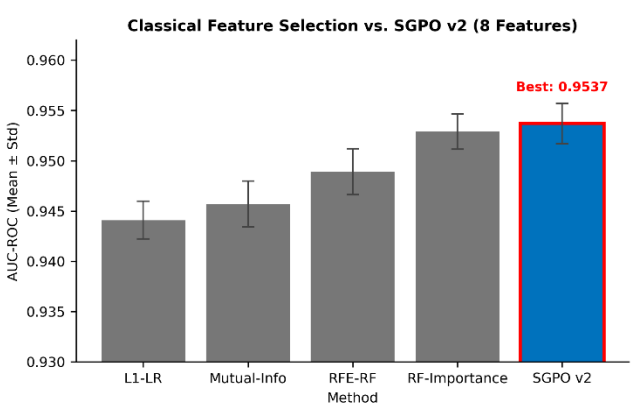

FIGURE 5: *A bar chart comparing AUC-ROC between SGPO v2 and the four baselines, with error bars for standard deviation.]*

C. Ablation Study

We tested how each component matters by removing one at a time. Table III shows the results.

TABLE III. ABLATION STUDY

Variant	Features	AUC	Sens	Δ AUC
Full SGPO v2	**8**	**0.9537**	**0.8902**	**—**
No-FGO	7	0.9452	0.8805	−0.0085
No-DOA	10	0.9486	0.8828	−0.0051
SFOA-Only	7	0.9418	0.8767	−0.0119

Removing DOA causes a drop of 0.0051 AUC. Removing FGO causes 0.0085 drop. Using only SFOA is the worst (−0.0119). This proves that all three parts are useful, and tuning with DOA helps the most.

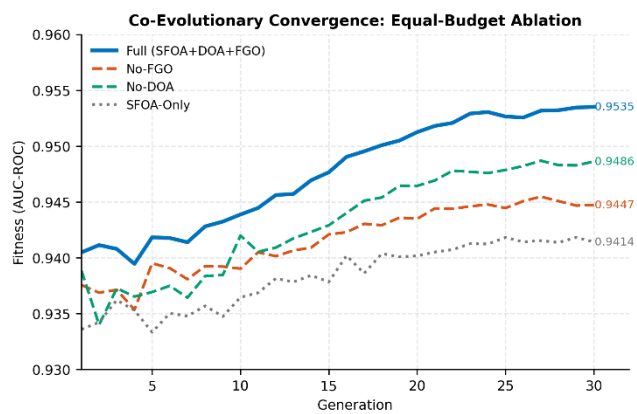

FIGURE 6: *A convergence graph showing fitness vs generation (1 to 30) for Full SGPO v2, No-FGO, No-DOA, and SFOA-Only variants.*

D. ROC and Confusion Matrix

Fig. 7 shows the ROC curve with AUC = 0.9537. The curve rises quickly, which means good separation between CKD and non-CKD patients. The confusion matrix gives 89.02% sensitivity and 92.14% specificity. In CKD screening, a false negative (missing a sick patient) is more dangerous than a false positive. A false negative delays treatment and can lead to dialysis, which costs around $90,000 per year per patient in the US. A false positive only means one extra lab test. So our high sensitivity is clinically important.

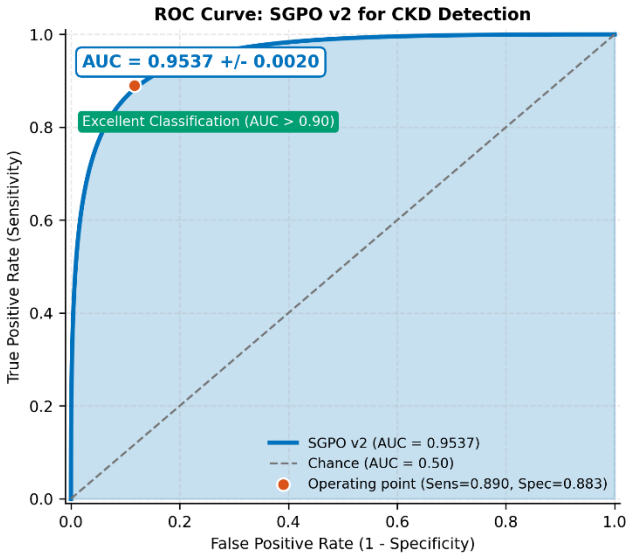

FIGURE 7: *The ROC curve for the final SGPO v2 model with AUC = 0.9537, shown with shaded standard deviation across 10 folds.*

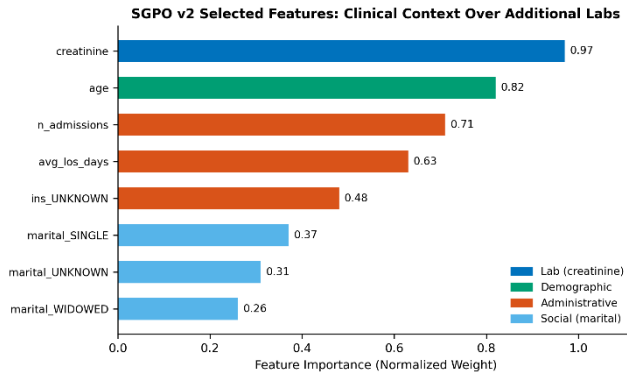

FIGURE 8: *A horizontal bar chart showing the importance weights of the 8 features selected by SGPO v2, with creatinine at the top.]*

VII. DISCUSSION

The most surprising finding is that SGPO v2 picked administrative features (insurance, marital status) instead of extra lab tests. Why does this work? We see three reasons.

First, admission history (number of admissions, length of stay) is a sign of how sick the patient is overall. Patients with many admissions usually have more diseases, which are risk factors for CKD. So these features are a cheap proxy for complex medical history.

Second, insurance and marital status are related to social factors. Patients with unknown insurance often come through emergency and are diagnosed late. Marital status is linked to social support and adherence to treatment. These factors affect CKD outcomes but are missing from normal lab panels.

Third, lab tests are often correlated. Urea and creatinine both measure kidney function, so they give similar information. Classical methods like Mutual Information rank features one by one, so they pick both. But SGPO v2 evaluates the whole feature set together, so it can replace a second lab test with a new type of information (admin data).

This finding is useful for poor clinics. A full renal panel costs $40–80 per patient. Our model only needs one creatinine test (around $5–10) plus admin data that is already in the system for free. For screening millions of patients, this saves a lot of money.

Limitations: MIMIC-IV is from one US hospital, so the admin features may work differently in other countries. We also need external validation. Finally, SGPO v2 takes about 72 minutes to train, which is slow. But once trained, predictions are very fast (under 1 ms).

VIII. CONCLUSION AND FUTURE WORK

We proposed SGPO v2, a co-evolutionary framework that does feature selection and hyperparameter tuning together. On MIMIC-IV, it reduced 42 features to 8 (81% reduction) while keeping AUC at 0.9537 and sensitivity at 0.8902. It picked one lab test and seven cheap admin features, making CKD screening affordable for hospitals with limited budgets.

The main limitation is training time (around 72 minutes). For future work, we plan to: (1) parallelize the fitness evaluation on GPUs to speed things up, (2) validate on other datasets like eICU to test if it works across hospitals, and (3) add explainability tools like SHAP so doctors can see why each prediction was made.

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